A 2016 Narrative Review of Autosomal-Recessive Inherited Retinal Degeneration

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1. ABCA4 allelic series — section 1

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Founder alleles complicate carrier-frequency interpretation in population isolates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 75329137, 29958938, 80801686.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 75329137, 32868928, 872348.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 79254663, 12575662, 32868928.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 3561697, 90825167, 3561697.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 37338224, 56629488, 90825167.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 99529323, 79254663, 3356986.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 21429210, 31227316, 11668832.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 98847593, 56629488, 872348.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 79254663, 98847593, 60291917.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 90825167, 87226112, 56722444.

2. Ophthalmic phenotyping — section 2

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 872348, 85822512, 18728563.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 87226112, 90825167, 73140907.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 29345192, 60291917, 56722444.

Founder alleles complicate carrier-frequency interpretation in population isolates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 56722444, 31227316, 29587139.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 99529323, 56629488, 21429210.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 98847593, 56307097, 872348.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 37338224, 73197957, 29345192.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 18728563, 56629488, 79254663.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 85822512, 36913910, 99529323.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 60291917, 56629488, 3999415.

3. PDE6B and rod-photoreceptor death — section 3

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Founder alleles complicate carrier-frequency interpretation in population isolates. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 56629488, 56307097, 96102625.

Founder alleles complicate carrier-frequency interpretation in population isolates. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 87226112, 96102625, 73140907.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 29958938, 56629488, 73197957.

Founder alleles complicate carrier-frequency interpretation in population isolates. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 99410841, 3999415, 73140907.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 37338224, 85822512, 32868928.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 11668832, 26687637, 99529323.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 73140907, 12575662, 87226112.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Founder alleles complicate carrier-frequency interpretation in population isolates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 11668832, 99529323, 93702783.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 93702783, 56307097, 60291917.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 99410841, 36913910, 56722444.

Table D3. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	9221552	3956066	0.4040
2	76560539	4140808	0.5788
3	5434035	1385453	0.4194
4	78439168	9492934	0.5227
5	35099379	3436900	0.6694
6	42269044	4014485	0.2661
7	17666185	5043115	0.4573
8	9836572	166287	0.4584

4. ARL6/BBS3 phenotypic spectrum — section 4

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Founder alleles complicate carrier-frequency interpretation in population isolates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 98847593, 80801686, 36913910.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 37338224, 29345192, 45667751.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 98847593, 4265899, 18728563.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 56722444, 11668832, 67827738.

Founder alleles complicate carrier-frequency interpretation in population isolates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 99529323, 32868928, 87226112.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 4265899, 90825167, 94130344.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 36913910, 13756769, 37338224.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 98847593, 3356986, 29345192.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Founder alleles complicate carrier-frequency interpretation in population isolates. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 29958938, 80801686, 872348.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 96102625, 14942703, 99410841.

5. DHDDS and arRP59 — section 5

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 29958938, 75329137, 3356986.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 29345192, 79254663, 79254663.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Founder alleles complicate carrier-frequency interpretation in population isolates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 36913910, 3999415, 80801686.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 18728563, 4265899, 99410841.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 80801686, 31227316, 87226112.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 4265899, 3356986, 87226112.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Founder alleles complicate carrier-frequency interpretation in population isolates. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 87226112, 36913910, 67827738.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 96102625, 31227316, 26687637.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Founder alleles complicate carrier-frequency interpretation in population isolates. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 75329137, 99410841, 45667751.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 12575662, 11668832, 87226112.

6. ARL6/BBS3 phenotypic spectrum — section 6

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 11668832, 60291917, 56307097.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 32868928, 56629488, 29345192.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 56629488, 56307097, 45667751.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Founder alleles complicate carrier-frequency interpretation in population isolates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 29587139, 26687637, 56629488.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 99410841, 79090001, 73140907.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 98847593, 73140907, 79090001.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 872348, 94130344, 90825167.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 79090001, 29587139, 56629488.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 3999415, 67827738, 31227316.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 75329137, 13756769, 11668832.

Table D6. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	8720650	6970328	0.4078
2	72928034	7826912	0.4159
3	27860841	7058838	0.3897
4	78493811	337691	0.8560
5	77367850	6391734	0.4770
6	47312267	5019888	0.7530
7	56340084	9039842	0.7470
8	73400637	3710025	0.4883

7. ABCA4 allelic series — section 7

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 26687637, 90825167, 73140907.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 32868928, 87226112, 98847593.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 31227316, 11668832, 73140907.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 4265899, 32868928, 56307097.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 36913910, 3356986, 14942703.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 98847593, 56307097, 18728563.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Founder alleles complicate carrier-frequency interpretation in population isolates. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 90825167, 56722444, 56722444.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 29345192, 21429210, 75329137.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Founder alleles complicate carrier-frequency interpretation in population isolates. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 29345192, 56722444, 18728563.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 36913910, 60291917, 67827738.

8. Autosomal-recessive inherited retinal degeneration — section 8

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 73140907, 13756769, 87226112.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 87226112, 93702783, 3999415.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 56629488, 73140907, 21429210.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 31227316, 99410841, 29587139.

Founder alleles complicate carrier-frequency interpretation in population isolates. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 56722444, 3999415, 872348.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Founder alleles complicate carrier-frequency interpretation in population isolates. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 96102625, 29587139, 872348.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 32868928, 12575662, 79254663.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 872348, 79090001, 80801686.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 3999415, 29345192, 4265899.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Founder alleles complicate carrier-frequency interpretation in population isolates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 37338224, 99410841, 73197957.

9. Variant filtering strategies — section 9

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 73197957, 12575662, 79254663.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 3999415, 3356986, 99529323.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 85822512, 21429210, 12575662.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 99529323, 4265899, 56307097.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 21429210, 99529323, 13756769.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 18728563, 872348, 96102625.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 94130344, 29345192, 93702783.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 11668832, 4265899, 32868928.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 85822512, 96102625, 94130344.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 12575662, 12575662, 73140907.

Table D9. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	9653585	3926201	0.9231
2	83993865	9905027	0.6609
3	26650845	7142648	0.1155
4	30273413	2509627	0.9076
5	19190631	1208058	0.0605
6	41386958	9993182	0.7485
7	76497676	4852082	0.4393
8	63007576	5111803	0.6993

10. ABCA4 allelic series — section 10

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 56722444, 3356986, 87226112.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 21429210, 93702783, 14942703.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 79090001, 94130344, 3999415.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 56307097, 80801686, 96102625.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 96102625, 29587139, 12575662.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 31227316, 4265899, 31227316.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 96102625, 99529323, 73197957.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 11668832, 3999415, 872348.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 14942703, 56722444, 56629488.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 32868928, 79254663, 18728563.

11. CERKL clinical heterogeneity — section 11

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 96102625, 56307097, 56307097.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 73197957, 56722444, 60291917.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 60291917, 3999415, 85822512.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 37338224, 56307097, 80801686.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 31227316, 73197957, 56629488.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 31227316, 56307097, 75329137.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 60291917, 93702783, 67827738.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 85822512, 13756769, 96102625.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 45667751, 26687637, 32868928.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 37338224, 3356986, 45667751.

12. Ophthalmic phenotyping — section 12

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 29958938, 29958938, 94130344.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 99529323, 12575662, 80801686.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 21429210, 99410841, 90825167.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 93702783, 13756769, 79254663.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 73140907, 12575662, 37338224.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 87226112, 4265899, 73140907.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 56629488, 75329137, 21429210.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 26687637, 3999415, 85822512.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 56722444, 80801686, 79254663.

Founder alleles complicate carrier-frequency interpretation in population isolates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 29345192, 94130344, 3356986.

Table D12. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	77889079	6131355	0.7319
2	12197496	4961131	0.3271
3	55867881	2956751	0.2014
4	72498083	6148070	0.5308
5	36678001	2770597	0.2575
6	64772520	4961651	0.7461

Index	Metric A	Metric B	Metric C
7	45561591	1942227	0.4684
8	10206509	2371510	0.7536

13. Variant filtering strategies — section 13

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 85822512, 4265899, 37338224.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 3561697, 29958938, 79254663.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 36913910, 73140907, 29345192.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 29345192, 29587139, 18728563.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 73140907, 3561697, 37338224.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 29587139, 45667751, 96102625.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 94130344, 56629488, 3561697.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 26687637, 3999415, 73197957.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Founder alleles complicate carrier-frequency interpretation in population isolates. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 99529323, 67827738, 56629488.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Founder alleles complicate carrier-frequency interpretation in population isolates. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 93702783, 11668832, 79254663.

14. RPE65-associated retinopathies — section 14

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 37338224, 4265899, 99410841.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 56629488, 93702783, 31227316.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 96102625, 79090001, 18728563.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 94130344, 60291917, 79254663.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 14942703, 99529323, 56307097.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 36913910, 32868928, 31227316.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 45667751, 93702783, 872348.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 60291917, 56722444, 87226112.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Founder alleles complicate carrier-frequency interpretation in population isolates. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 94130344, 79254663, 96102625.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 31227316, 87226112, 31227316.

End of distractor document. No content above answers any specific pedigree-level question; this document is included to test whether downstream agents correctly select the relevant source paper.